How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Abstract

What shapes the content of a recurrent navigation agent's learned internal state — the task, the architecture, or the structure of the visual input? Prior work has shown that blind PointNav agents spontaneously develop cognitive-map-like representations in their recurrent state under task pressure alone [Wijmans et al., 2023]. We show that the blind result is a special case of a broader principle: *the visual encoder's information capacity determines whether the LSTM has to carry world-frame spatial structure*. Holding task, optimiser, and LSTM backbone identical and varying only the visual pathway (no vision; a resolution-collapsed encoder; uniform vision; foveated vision at a fixed gaze; foveated vision with a learned gaze), we probe the recurrent state for world-frame position and heading. All five conditions encode top-layer GPS in their LSTM in the typical-episode window. What separates conditions is what happens *later*: when the encoder hands the LSTM minimal spatial-feature variety per step (blind: no encoder; matched-compute: 1×1 feature map), the LSTM *maintains* a persistent, integration-capable code across episode duration; when the encoder hands it richer spatial features (full-resolution encoders, 8×8 feature map), the LSTM *overwrites* the GPS code as those visual features take over. We call this the *encoder–memory race*; it recasts the "cognitive maps emerge without vision" story as "cognitive maps emerge — and persist — when the visual pathway cannot resolve the map on its own." A second finding cuts across conditions: five agents that solve the task equally well develop recurrent representations in essentially orthogonal linear subspaces, and cross-condition memory transplants incur task cost well above transplant mechanics — sensor structure shapes representational *format*, not only strength. A third question — whether static gaze location within a rich-encoder regime acts as a second content axis — is tested by a foveated-shifted causal control whose training is in progress. The H1 ordering reproduces on a held-out MP3D evaluation, consistent with an agent-internal rather than Gibson-specific mechanism. Three implications, each scoped to the recurrent PointNav setting we study. **(i)** Encoder capacity correlates inversely with linear, top-layer LSTM spatial encoding; *if* the correlation reflects causation (which the encoder-resolution scaling sweep, in progress, tests directly), it would also be a practical training-time lever for inducing cognitive-map-like internal representations. **(ii)** "More pixels, richer memory" is the wrong intuition within this setup; the relationship across our five conditions is inverse. **(iii)** The pattern parallels independent animal-navigation findings — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in the dark (with navigation nonetheless preserved), cross-modality hippocampal remapping in bats — without depending on their biophysics, suggestive of a sensor–memory coupling principle whose generality across architectures (transformers, non-RL learners, non-navigation tasks) we do not test here.

Preprint.

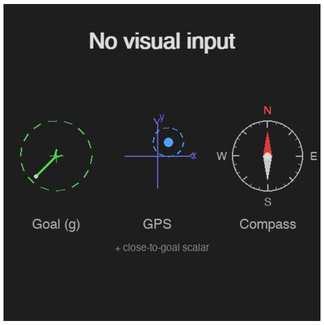
1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget, or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

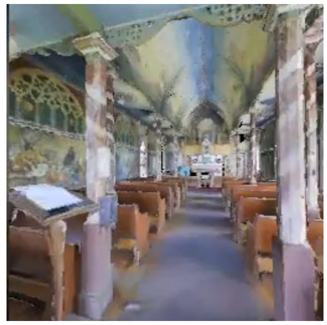
Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), encoder capacity (uniform's full ResNet-18 vs. matched-compute's 1×1-collapsed feature map), and gaze location (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We adopt the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is recurrent, and matching that architecture lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. We discuss in §5 whether the logic plausibly extends to transformer-backed or non-navigation systems — an empirical question we do not settle here.

We test three hypotheses. **H1 (encoder–memory race).** Agents whose visual encoder cannot resolve world-frame position or heading should develop these quantities in their recurrent state; agents with a rich visual encoder should not. Signature: current-state GPS / compass probe R^2 tracks visual-encoder information content monotonically across the five conditions, with persistent decoding at lags of ≥ 10 time-steps. **H2 (sensor structure $\rightarrow$ format divergence).** Different sensor conditions should produce internal representations in distinct linear subspaces, not scaled versions of a common code. Signature: pairwise CKA near zero, probe weights fail to transfer across conditions while within-condition probes succeed, and cross-condition memory transplants degrade task performance beyond same-condition self-transplants. **H3 (gaze location $\rightarrow$ orthogonal content lever).** Where a foveated sensor is centred should act as a representational-content axis independent of the sensor transform. We test this with a foveated-shifted control in which gaze is hardcoded to a different image location from the default central gaze; same foveation transform, same training budget, different gaze target. Our minimal learned-gaze module collapsed to a static output (Appendix E), so we use that collapsed location for the shifted-gaze control rather than as evidence about active gaze; the foveated-shifted training is in progress (§4.5). Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings H1 is supported as a *temporal-stability* claim, not an existence claim: all five conditions encode top-layer GPS in the typical-episode window, and only the bottleneck conditions (blind, matched-compute) maintain it across episode duration; rich-encoder conditions overwrite it as visual features accumulate. The pattern is preserved on a held-out MP3D evaluation. The encoder feature-map probe (§4.4) sharpens the mechanism: *none* of the three sighted encoders linearly decode GPS from their visual stream, so the H1 divergence is not about encoder-level position retention but about how much spatial-feature variety the encoder hands to the LSTM — and whether that variety is enough to substitute for integrating the GPS-sensor signal across time. H2 is supported by four convergent cross-condition divergence measures: pairwise CKA at the noise floor, probe-transfer failure to off-manifold values, 1-NN purity 1.000, and behaviourally costly cross-condition memory transplants whose 5×5 matrix is asymmetric (blind donations hurt rich-encoder agents far more than vice versa). H3 asks whether gaze location is a second content axis within the rich-encoder

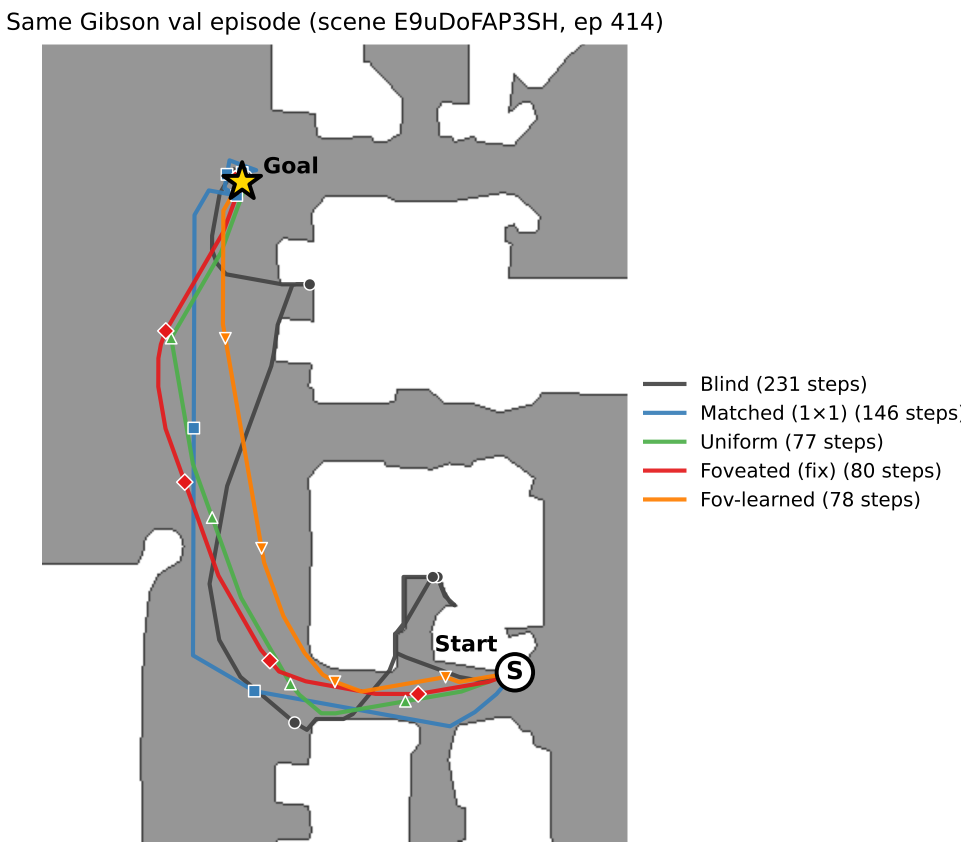


(a) Blind



(b) Uniform

(c) Foveated (fixed gaze)



(d) The 5 conditions on the same MP3D-train episode (scene E9uDoFAP3SH, ep 40422 in the env iterator; same paired index 414 across our probing data), trajectories overlaid on the Habitat-rendered top-down occupancy map. Each condition's SPL on this episode is within ± 0.10 of its per-condition median, so all five are typical successful runs. Bottleneck conditions (blind 231 steps, matched 146) take longer paths through the corridor; rich-encoder conditions (uniform / foveated / fov-learned, 77–80 steps) take near-direct paths along the same corridor.

Figure 1: The five visual conditions differ only in what reaches the agent's encoder, but produce visibly different navigation behaviours. *Inputs:* (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform: 256×256 full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). Two additional conditions not pictured: *matched-compute* (a down-sampled uniform view at 48×48, whose total pixel budget matches foveated and whose ResNet-18 encoder produces a 1×1 feature map) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). *Behaviour:* (d) all five conditions solve the same PointGoal navigation task; the trajectory shape and length differ systematically, previewing the encoder–memory race finding (§4.2).

regime; the clean causal test (foveated-shifted) is in training. A non-linear (MLP) probe at the top layer recovers position from rich-encoder networks ($R^2 \in [0.51, 0.73]$), so H1 is most precisely a claim about *linear, top-layer* GPS encoding (i.e. what the linear DD-PPO action head can use); we cannot from MLP recovery alone exclude exploitation of position-correlated features (wall distance, scene identifiers) rather than position itself.

Contributions (i) A controlled five-condition ablation that decouples "no vision" from "visual-encoder bottleneck" via the matched-compute condition, making the encoder–memory race claim falsifiable. (ii) Convergent evidence (linear probes, MLP probes, CKA, probe transfer, 1-NN purity, memory transplant, shortcut discovery, MP3D generalisation, per-layer probes) for two of the three hypotheses, with a clean methodology disclosure including the deterministic-rollout protocol that fixes a probe-collection confound documented in §3.3. (iii) A mechanistic re-reading of Wijmans et al.'s blind-agent finding: cognitive maps emerge not because vision is absent but because the visual encoder cannot resolve the map, predicting an encoder-capacity scaling curve we begin to test in Appendix H.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

What is new here Three things distinguish this work from the studies above. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the matched-compute condition that decouples "no vision" from "visual encoder bottleneck" — the lever that makes the encoder–memory race claim falsifiable. (ii) A protocol-corrected probing methodology: deterministic action selection at probe time (the policy used by every other component of our eval pipeline) rather than stochastic sampling, which we found inflates probe R^2 for high-entropy policies via a 4-step quasi-static-trajectory artefact (§3.3, Sampling protocol). (iii) Convergent behavioural validation of the probe-based H1 and H2 findings via memory transplant and shortcut discovery, both of which sit *outside* the linear-probe framework.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix F.

3.2 Visual Conditions

Five conditions vary only the visual input (Figure 1):

1. *Blind:* no visual input; non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. *Uniform:* full-resolution 256×256 RGB encoded by ResNet-18.
3. *Foveated (fixed):* eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre; same ResNet-18 encoder.
4. *Matched-compute:* $48{\times}48$ uniform RGB. At this resolution, ResNet-18's 32-fold spatial downsampling collapses the feature map to $1{\times}1$, removing the spatial dimensions of the encoder's output (channel information is preserved as a flat 512-d vector). **[TODO: Verify whether channel info alone in matched-compute encodes any usable position signal — a small probe on the encoder feature map (not LSTM state) would settle this.]**
5. *Foveated (learned):* same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

Matched-compute is central to H1: its $1{\times}1$ encoder feature map separates the "no vision" account from the "encoder bottleneck" account. The $48{\times}48$ input also has lower spatial resolution than uniform's $256{\times}256$, so this single condition cannot fully separate the $1{\times}1$ collapse from low input resolution; the resolution scaling sweep (Appendix H) tests this directly. The foveated conditions (fixed, shifted, learned) disable the running-mean-and-variance input normaliser used by uniform / matched, to avoid an in-place autograd conflict along the gaze-decoder gradient path; **[TODO: normaliser-invariance control (F2) is in training.]**

3.3 Probing Methodology

After training, we freeze all weights and collect per-step top-layer LSTM hidden states (512-d) paired with ground-truth labels from $n{=}500$ Gibson held-out episodes under *deterministic* argmax-action rollouts (matching our shortcut, transplant, and standard eval protocol; the alternative stochastic protocol gives trivial ~ 4-step rollouts and is documented in Appendix A). We train Ridge regression probes ($\alpha = 10$) with episode-level 5-fold cross-validation, report mean $\pm\sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019]. Probed variables: GPS position, compass heading, distance-to-goal, lag-k path history ($k \in \{0, \ldots, 20\}$), visited-region occupancy, full occupancy grid, per-unit spatial information, and the ego-frame goal vector. For foveated-fixed, all analyses use ckpt.36 at $\sim$174M frames (the last clean intermediate before the silent-corruption window, Appendix F).

3.4 Cross-condition + behavioural probes

For H2, *linear CKA* [Kornblith et al., 2019] measures representational similarity ($1 =$ identical, $0 =$ unrelated); *probe transfer* trains a GPS probe on one condition and evaluates on another (catastrophic failure indicates different linear formats). For behavioural validation: *memory transplant* pastes a donor's mid-episode hidden state into the recipient at step 30 and compares baseline / self-transplant / cross-transplant SPL across 150 episodes per pair; *shortcut discovery* runs paired same-scene-different-goal episodes (20 scenes $\times$ 10 pairs) and compares second-episode SPL under reset vs. persistent LSTM. A cross-heading probe was implemented but returned "insufficient samples" at our probe sizes (Appendix A).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers this section returns to. All sighted conditions solve PointGoal at 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. Success is tightly bunched, so "the agents differ in what they do" is an unlikely explanation for the encoded-content gap we report (it does not strictly rule it out). The probe machinery itself is calibrated: distance-to-goal decodes robustly across all five conditions ($R^2 \in [0.81, 0.90]$; Appendix D for the companion plot), confirming that Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them; and Hewitt–Liang label-permutation selectivity is high in the conditions where the GPS code itself is high (blind GPS / compass selectivity $+0.95/+0.81$; matched $+0.72/+0.56$), consistent with a genuine signal rather than label memorisation. Two training-stability caveats are disclosed in full in Appendix F: foveated-fixed uses ckpt.36 at $\sim$174M frames (last clean checkpoint before a silent NaN-gradient corruption at $\approx$182M); and matched-compute's 1×1 feature-map collapse, originally noticed as a training quirk, is what makes matched-compute a useful test of the encoder-bottleneck framing (we cannot from one condition alone fully separate the 1×1 feature-map effect from the lower input resolution; see §3.2 and §4.4).

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind $>$ matched $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	**+0.95±0.02**	**+0.81±0.08**
Matched-compute	250M	0.67	0.98	**+0.78±0.10**	**+0.64±0.10**
Uniform	250M	**0.85**	**0.99**	−0.31±0.86	+0.36±0.23
Foveated (fix)	174M	0.78	0.96	+0.06±0.88	+0.07±0.69
Fov-learned	250M	0.82	0.98	−2.43±3.98	−1.34±3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

Finding Visual-encoder capacity inversely tracks LSTM spatial encoding across our five conditions: bottleneck conditions (blind, matched) encode GPS / compass at the top layer; rich-encoder conditions

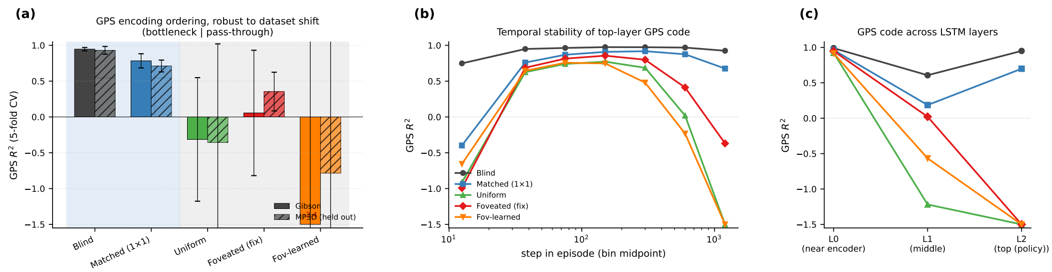


Figure 2: H1 evidence in three panels. *(a)* GPS R^2 (5-fold CV, Ridge $\alpha = 10$, full-length deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition; same checkpoints, no re-training). Blind and matched-compute (bottleneck conditions, blue background) decode world-frame GPS robustly on both datasets; uniform / foveated / foveated-learned (rich-encoder, grey background) do not, except for foveated recovering to a mild positive ($R^2 = +0.35$) on MP3D. The Compass and DtG numbers are in Table 1 (compass shows the same ordering at smaller magnitude; DtG decodes everywhere because every agent must track it; Appendix D for the companion bar plots). *(b)* Top-layer GPS R^2 binned by step-in-episode (out-of-fold prediction): all five conditions encode top-layer GPS in the typical-episode window (steps $\sim$ 25–200); the rich-encoder conditions then *overwrite* the code (negative R^2 in the long-tail $800+$ bin), while bottleneck conditions hold $R^2 > 0.7$ to step $800+$. The headline "rich-encoder $\approx$ chance" in panel (a) is therefore primarily long-tail behaviour. *(c)* Per-LSTM-layer GPS R^2 (single-fit on top-layer hidden states): all five conditions have GPS at Layer 0 (where the GPS sensor embedding enters the LSTM input vector — this lower-bound is at least partly a sensor-pass-through effect, not the network "computing" position); the divergence is at Layer 2, the policy readout.

(uniform, foveated, foveated-learned) decode at chance (Figure 2a, Table 1). DtG decodes robustly in every condition because every agent must track it. **[TODO: Whether the correlation reflects causation is what the §4.4 experiments and the Appendix H resolution sweep test; multi-seed gap-fills are in progress.]**

Temporal stability is the precise claim The no-cap probe conflates "does the LSTM ever encode GPS" with "does it maintain the encoding across episode duration." Step-binned out-of-fold probing separates them (Figure 2b): all five conditions encode top-layer GPS in the typical-episode window; rich-encoder conditions then *overwrite* the code (chance by step 400–800, deeply negative in the long-tail bin), while bottleneck conditions hold it stable to step 800+. The Table 1 "rich-encoder $\approx$ chance" headline is therefore long-tail behaviour, not mid-episode. *H1 is most precisely a temporal-stability claim about the linearly-decodable top-layer GPS code.* A lag-k probe corroborates: blind GPS decodes at $R^2 = 0.92$ at $k = 20$, matched > 0.5; rich-encoder conditions are at chance for every $k \leq 10$.

Layer specificity + non-linear probe All five conditions have GPS at LSTM Layer 0 because the GPS-sensor embedding enters there (Figure 2c). The H1 distinction is what happens through Layers 1–2: bottleneck retain to Layer 2, rich-encoder discard. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, this maps to "policy-accessible". A 2-layer MLP on the same top-layer states recovers GPS for rich-encoder conditions ($R^2 \in [0.51, 0.73]$); H1 is therefore precisely about *linear* top-layer decodability, not whether *any* probe can recover position. The MLP result alone cannot exclude exploitation of position-correlated features (wall distance, scene identifiers); Appendix B discusses both. The bottleneck-vs-pass-through partition is preserved on held-out MP3D (Figure 2a); two single-seed cross-dataset shifts (foveated GPS chance $\rightarrow +0.35$; foveated-learned compass $-1.34 \rightarrow +0.41$) await multi-seed replication.

Proposed mechanism We *propose* the following account, consistent with the data but not directly tested by a causal intervention. In PointNav, GPS and compass enter the LSTM input at Layer 0 directly, via a 32-d sensor embedding concatenated with the visual encoder's flattened output (Figure 2c shows GPS at Layer 0 across all five conditions, $R^2 > 0.91$). The encoder feature-map probe (§4.4, Figure 4) shows that none of the three sighted encoders linearly re-derive GPS from their visual stream: matched $R^2 = -3.14$, uniform -0.32, foveated -0.65. The encoder–memory race is therefore not about whether the encoder *re-derives* position; it is about how much *visual* *feature variety* the encoder hands to the LSTM, and whether that variety is enough to substitute for integrating the Layer-0 GPS-sensor signal across time. Bottleneck conditions (blind, no encoder; matched-compute, 1×1 spatial output $\rightarrow$ minimal feature variety per step) leave the LSTM with little visual structure to substitute, so it integrates the Layer-0 GPS signal across time and *maintains* a top-layer linearly-readable GPS code. Rich-encoder conditions (uniform / foveated / foveated-learned, 8×8 spatial output $\rightarrow$ many position-correlated visual features) give the LSTM enough visual variety to base actions on those features instead, and the top-layer GPS code drifts off as the episode progresses. Task success is tightly bunched, so a skill-gap artefact is unlikely to explain the encoding gap. The temporal dynamic in Figure 2b (rich-encoder GPS code emerging early at Layer 0 then decaying through Layers 1–2 to chance at the top layer) is the direct fingerprint of this account. **[TODO: A direct causal test of the mechanism — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.]**

Relation to Wijmans et al. Wijmans et al. [Wijmans et al., 2023] showed blind agents develop spatial representations; our blind result ($R^2 = 0.95$) replicates this and matched-compute extends it to a condition that has visual input but 1×1 spatial encoder output. The shared trigger is therefore minimal visual-feature variety per step, not absence of vision. We discuss the parallel to animal-navigation findings (compensatory hippocampal coupling in congenitally blind humans, grid-cell collapse under darkness, cross-modality remapping in bats) in §5.3 as convergent evidence, not as a mechanistic claim.

4.3 Format-level divergence across conditions (H2)

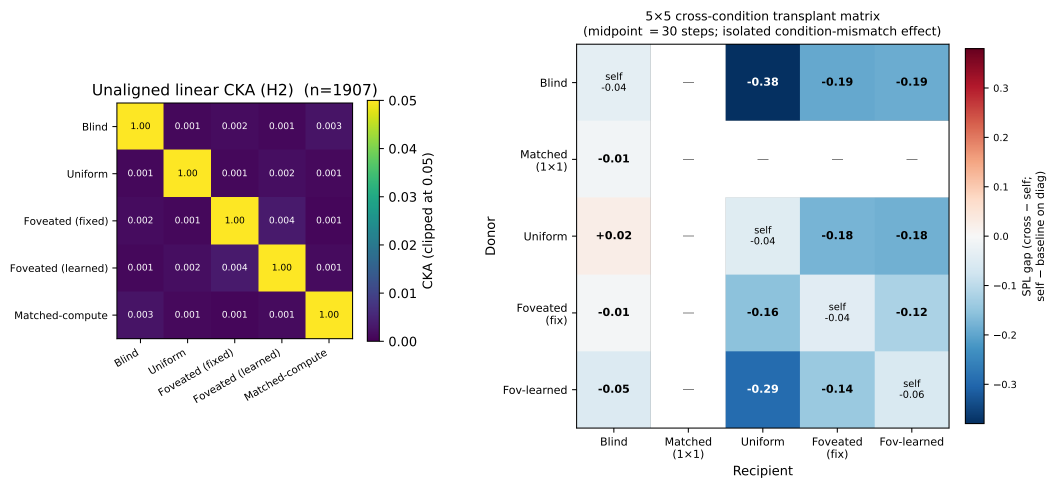


Figure 3: H2 convergent evidence, two views. *Left:* Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. *Right:* 5×5 cross-condition memory-transplant matrix at midpoint t=30. Cells off-diagonal report cross-transplant SPL minus the recipient's self-transplant SPL (the isolated condition-mismatch component, with rollout-divergence mechanics subtracted); diagonal cells report self-transplant SPL minus baseline (the rollout-divergence noise floor, ≈ -0.04 to -0.06). The matrix is asymmetric: blind→uniform incurs -0.38 SPL while uniform→blind is essentially neutral ($+0.02$); foveated-learned→uniform -0.29, blind→foveated -0.19. *Pairs that are indistinguishable to linear similarity (CKA noise floor) differ markedly under transplant.* Cells with '—' are not yet measured (matched as recipient, jobs in flight). A midpoint-sweep view of three representative pairs across midpoints $\{0, 15, 30, 60, 90\}$ confirming the noise-floor and stability arguments is in Appendix D (Figure 8).

Behavioural test (memory transplant) The most direct test of "do these representations matter?" is to paste a donor's mid-episode hidden state into a recipient's policy at step 30 and let the recipient drive the rest (Figure 3, right). The isolated condition-mismatch cost — cross-transplant SPL minus recipient's self-transplant SPL — shows three patterns. *Asymmetry.* Blind→uniform incurs -0.38 SPL while uniform→blind is neutral within self-noise; bottleneck-donor states hurt rich-encoder

Table 2: Probe-transfer GPS R^2 on deterministic-rollout hidden states: train on row condition, test on column. Self-accuracy diagonal is 0.89–0.99; every cross-condition transfer is catastrophically negative ($R^2 \ll -800$, often $\ll -10^4$), reflecting that a probe trained on one condition's hidden-state geometry does not just degrade on another — it predicts off-manifold values.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	**+0.99**	−844	−11,301	−10,731	−55,706
Uniform	−1,383	**+0.90**	−7,046	−3,305	−110,317
Fov-fix	−2,091	−2,792	**+0.92**	−1,263	−39,100
Fov-lrn	−6,722	−2,423	−6,205	**+0.89**	−38,842
Matched-compute	−10,621	−4,832	−4,984	−58,169	**+0.96**

recipients far more than the reverse. *Recipient ranking.* Uniform suffers most from *any* foreign donor (column mean ~ -0.20); blind least (~ -0.02 to -0.05). Blind tolerates foreign donors; rich-encoder recipients do not. *Order-of-magnitude probe-similarity blind spot.* Pairs that are indistinguishable under linear similarity (CKA $< 10^{-4}$ across the board) differ by 0.01–0.38 SPL under transplant. **[TODO: 4 matched-recipient cells of the 5×5 are queued; asymmetry will be tested for completeness on landing.]**

Convergent linear-similarity evidence Three further measures point to the same "different format" conclusion. *Pairwise CKA* [Kornblith et al., 2019] at the noise floor across all condition pairs (Figure 3, left; within-condition split-half CKA ≈ 1, so the near-zero off-diagonals are not estimator noise). *Cross-condition probe transfer* (Table 2) is catastrophically negative in every off-diagonal cell, so a probe trained on one condition's hidden states predicts off-manifold values on another. *1-NN purity*: among $1500\times5 = 7500$ pooled top-layer states, every sample's Euclidean nearest neighbour shares its condition (purity 1.000 vs. chance 0.20); a t-SNE projection (Appendix D) confirms this visually. **[TODO: 1-NN at larger sample size and under domain shift (MP3D / fov-shifted) is a further-investigate item.]**

Ruling out alternatives Probe capacity is not the bottleneck (self-probes succeed everywhere). Sample-size asymmetry is ruled out by common-sample truncation. Task competence is bunched (96–99%), so the divergence is not a skill difference. Probe transfer fails symmetrically, ruling out one condition's code simply embedding another's.

Boundaries Our similarity tests are linear or near-linear; non-linear measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (sharp fovea + coarse periphery, depth-only) might produce intermediate geometries.

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

Foveated-fixed and uniform fall in the same rich-encoder pass-through regime on the headline H1 metric (top-layer GPS R^2 at chance), but they are not interchangeable on H2 / behavioural / dataset-shift axes (Table 3). The take-away: *within the rich-encoder regime defined by H1, foveation and uniform produce internally distinct representations whose differences only the H2-and-beyond machinery makes visible*.

Table 3: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 3, Figure 5, Figure 4. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, *indistinguishable*
Encoder $\to$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	**differ by 0.71**
Shortcut SPL drop %	21%	41%	**uniform $2\times$ more brittle**
Memory-transplant cost (fov $\to$ uni)	−0.10 SPL beyond self-noise		**not interchangeable**
Pairwise CKA(fov, uni)	$< 10^{-4}$		**near-orthogonal subspaces**
1-NN purity (pooled)	1.000 vs. chance 0.20		**linearly separable**

The four in-flight experiments below test *which* aspect of foveation drives these differences — sensor transform strength, spatial-sampling vs. blur, gaze location, or encoder feature-map representation. Numbers will populate the sub-paragraphs in revision; current status in Appendix G.

Foveation strength sweep (F1–F4, in progress) We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ (alongside the existing $\sigma_{\max} = 8$), holding everything else fixed — gaze location, encoder stack, training budget, training pool. The expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform (encoder still resolves world-frame); at high $\sigma_{\max}$ it should approach matched-compute (encoder cannot resolve world-frame, LSTM compensates). The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever, not a binary one.

Spatial sampling vs. blur (F3 log-polar control, in training) Gaussian blur removes high-frequency content but preserves *spatial layout*: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples the image with a non-uniform retinal grid (e.g. 64×64): the encoder's spatial output drops to $\sim 2 \times 2$, much closer to matched-compute's 1×1 than to uniform's 8×8. This is the architectural commitment "foveated vision" makes in biology, and it isolates the variable our sharpened H1 mechanism (§4.4, encoder feature-map paragraph) identifies as the trigger: *spatial-feature variety in the encoder output*.

Predictions, falsifiable. If the encoder–memory race is driven by encoder spatial output dimensionality (not by "bottleneck of any kind"), then log-polar should produce an LSTM GPS R^2 *between* matched-compute (current $+0.78$) and uniform (current ≈ 0), likely ≥ 0.3, with corresponding behavioural shifts (longer paths than uniform, larger shortcut SPL drop closer to blind / matched). If instead log-polar matches uniform's near-chance H1 readout, the mechanism is not pure encoder-output-dimensionality and we need to reframe (e.g. allow the encoder to learn task-useful features from any spatial layout, rendering "bandwidth" the wrong axis). Either outcome is informative; F3 is in training (~14M / 250M frames at submission, ETA 2–3 days). Numbers will populate this paragraph in revision; the prediction was written before the result was available.

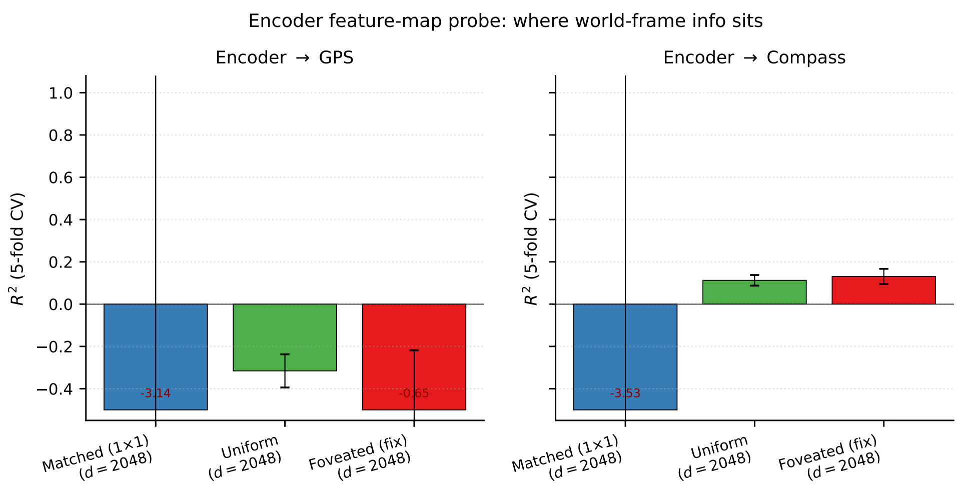


Figure 4: Encoder feature-map probe (post ResNet-18, pre-LSTM, 5-fold CV Ridge). Encoder dim $d = 2048$ (compressed flatten) for all three sighted conditions. None of the three encoders linearly decode GPS (matched $R^2 = -3.14$, uniform -0.32, foveated -0.65); the encoder–memory race is therefore not about encoder-level GPS information availability. Compass shows a small positive on uniform / foveated ($+0.11, +0.13$); matched compass is also chance.

Encoder feature-map probe: where world-frame info actually sits Probing the visual encoder's output directly (post ResNet-18, pre-LSTM, via forward-hook on the compression layer) gives a striking result: *none of the three sighted encoders linearly decode GPS* (Figure 4). Compass also fails or is well below ceiling. *This sharpens the H1 mechanism*: the divergence happens inside the LSTM (or at the GPS-sensor pass-through entering Layer 0), not at the visual encoder. Bottleneck conditions integrate the GPS-sensor input across time because their visual stream gives the LSTM minimal feature variety to substitute; rich-encoder conditions have richer (though still position-blind) visual features that the LSTM uses instead, overwriting the Layer-0 GPS code as the episode progresses. Foveated does not differ from uniform on this measure (overlapping uncertainty), so its pass-through behaviour is consistent with "rich-encoder" rather than a foveation-specific effect at the encoder level.

Foveated-shifted (gaze location, in progress) Gaze location is treated separately as H3 (§4.5), but the foveated-shifted control directly informs the foveation slot too: training a foveated agent with hardcoded static gaze at $(0.49, 0.62)$ matches foveated-learned's collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated-learned's specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction and not to the static gaze location alone.

Why this slot exists Our title and framing focus on foveated vision; the present five conditions place foveated-fixed and foveated-learned within a framework but do not isolate foveation-specific effects (under simple Gaussian blur both behave like uniform). The four experiments above target specific foveation-specific predictions of the encoder–memory race framework. Each is a falsifiable test, designed and submitted at the time of writing. None depends on the others to be informative.

What we already conclude From existing data (Table 3), foveation under $\sigma_{\max} = 8$ Gaussian blur is *not a copy of uniform*: it converges with uniform on H1 but diverges on H2 (subspace orthogonality, 1-NN separability), behavioural memory reliance (shortcut drop), transplant compatibility, and dataset-shift recovery. The encoder feature-map probe further shows the encoder discards linear position regardless of condition — so *whatever foveation does differently from uniform, it does it inside the LSTM in subspaces invisible to a linear-GPS probe*. The in-flight experiments below probe *what* that "something different" is.

4.5 Gaze location as a second axis (H3)

H1 and H2 identify sensor structure as one axis on representational content. H3 asks whether gaze *location* is a second, dissociable axis. We test the gaze-location form (two foveated agents with gaze centred at different static positions should produce different content); the active-gaze form is out of scope of our minimal gaze decoder.

The learned-gaze module collapses Our minimal learned-gaze implementation (deterministic sigmoid MLP, slow updates per 128-step segment; Appendix E) freezes at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ after 250M frames; anti-collapse regularisers restore variance only at task-cost. We do not claim active gaze cannot be learned — our decoder is minimal by design — but we use the collapsed gaze position as the empirical anchor for the gaze-location test: the trained agent is *effectively foveated with a static gaze* 30 *pixels below image centre*.

Gaze-location test (foveated-shifted, in progress) Both rich-encoder foveated variants sit in the pass-through regime under current-state probing (Table 1), so the H3 test is whether *within* the pass-through regime, gaze location shifts content along other axes — non-spatial feature usage, behavioural memory-reliance (shortcut, Figure 5), or transplant compatibility. To isolate gaze location from training-dynamics confounds, we train a foveated-shifted control: an agent identical to foveated-fixed except gaze is hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$ — matching foveated-learned's collapsed gaze position but with no learned-gaze module in the optimisation path. **[TODO: Training in progress at submission; results will populate this paragraph.]**

Shortcut: a 2×2 dissociation between probe-readable and policy-used memory Plotting probe-readable GPS R^2 against behavioural memory reliance (shortcut SPL drop) reveals a 2×2 pattern (Figure 5). Blind sits in the "has GPS, uses memory" diagonal; foveated and foveated-learned in the "no GPS, ignores memory" diagonal — both consistent with H1. The off-diagonal cells are the interesting ones: *matched-compute has GPS but ignores memory* (linear top-layer cognitive map, low policy weight on it), and *uniform has no linear GPS but heavy memory reliance* (integrates non-spatial scene-history / visual-landmark features). "Probe-readable" and "policy-relied-on" are therefore distinct constructs that can dissociate in either direction.

Figure 6 shows the failure modes qualitatively differ across regimes: blind's persistent trajectory clusters around the previous-episode goal (literal "locked onto old position"); uniform and foveated wander diffusely ("locked onto old visual context" would fit). The visualisation cannot itself dis-

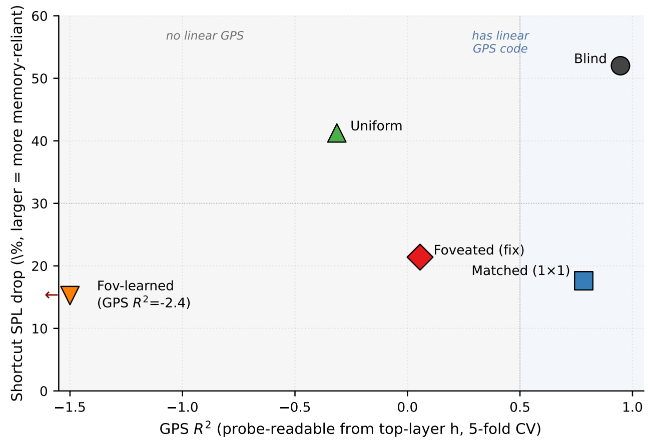


Figure 5: Probe-readable GPS (H1) vs. behavioural memory reliance (shortcut). The shortcut experiment runs paired episodes in the same scene with fresh goals and compares second-episode SPL under *reset* (LSTM zeroed) vs. *persistent* memory; larger drop = more memory-reliant policy. *x-axis:* GPS R^2 (5-fold CV mean) from Figure 2a. *y-axis:* relative shortcut SPL drop (%). 20 Gibson scenes $\times$ 10 paired episodes per condition. Fov-learned's $R^2 = -2.43$ is clipped at -1.5 (arrow). The 5 conditions populate a 2×2 dissociation: blind "has GPS code, uses memory"; foveated-fixed and foveated-learned "no linear GPS, ignores memory"; **matched** and **uniform** are the two opposite anomalies (matched: "has GPS but ignores memory"; uniform: "no linear GPS yet still memory-reliant").

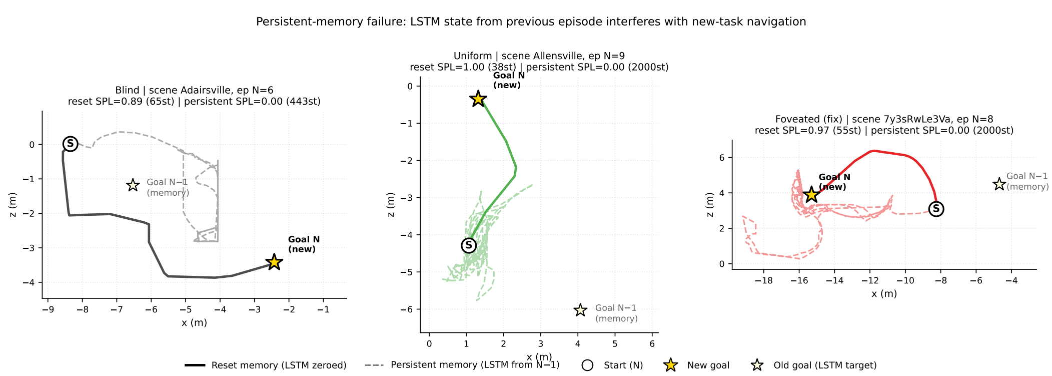


Figure 6: Persistent-memory failure made visual: trajectories from one paired-episode run per condition where reset memory succeeds and persistent memory fails. Same scene, same start position, same goal between the two memory conditions; the only difference is whether the LSTM is zeroed (reset) or carries over from the previous episode (persistent). Blind (left) shows the cleanest interpretation of the shortcut SPL drop: persistent agent oscillates around the *old* goal location (light star, $N-1$) instead of going to the new one (gold star, N), consistent with the "LSTM still encodes previous task" reading. Uniform (centre) and foveated (right) persistent runs wander more diffusely without obviously locking onto the old goal — consistent with their hidden state encoding scene-history / visual features rather than a position-readable memory of the previous task. The contrast in trajectory length is also informative (blind reset 65 steps vs. persistent 443; uniform 38 vs. 2000 max; foveated 55 vs. 2000).

tinguish position-locked from context-locked failure but is consistent with the 2×2 dissociation. **[TODO: Both anomalies rest on single-seed runs; multi-seed replication is in progress and would promote the dissociation from candidate to robust. A direct "policy GPS reliance" test (ablating the GPS sensor mid-rollout) would distinguish "policy reads the GPS code" from "policy reads non-GPS memory" — the matched anomaly predicts a small SPL drop, the uniform anomaly a large one.]**

4.6 Boundaries and additional probes

Three probes that do *not* discriminate define the boundary of our claims. Coarse occupancy decodes above chance everywhere; metric 20×20 occupancy fails everywhere (consistent with [Ramakrishnan et al., 2025] that high-resolution metric layout needs non-linear probes); the ego-frame goal vector is at chance everywhere because PointGoal supplies it as a per-step input. Population coding (Appendix D) sharpens H1 in a direction we did not expect: rich-encoder conditions have a few units with *higher* peak spatial information than blind, but those peaked units do not decode GPS — suggesting they encode features merely correlated with position (visual landmarks, trajectory phase). Blind's broader, lower-peak distribution carries the actual linear position code.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

Our findings unify under a candidate account we call the *encoder–memory race*: top-layer linear GPS encoding in the LSTM is *maintained* when the encoder hands the LSTM minimal spatial-feature variety per step (blind, matched-compute), and *overwritten* when richer visual features substitute (uniform, foveated, foveated-learned). The encoder feature-map probe (§4.4) confirms that no sighted encoder linearly re-derives GPS, so the H1 race is not encoder-vs-LSTM at the position-information level; it is at whether the LSTM has visual variety enough to substitute for integrating its Layer-0 GPS-sensor signal across time. H2 (§4.3) cuts orthogonally: *what* the LSTM encodes lives in distinct linear subspaces across conditions, and the difference matters for behaviour (cross-condition memory transplants drop SPL beyond transplant-mechanics noise; the 5×5 matrix is asymmetric). H3 (§4.5) asks whether, within the rich-encoder pass-through regime, gaze location is a second content axis; the foveated-shifted causal control is in training. We present the unifying account as a candidate, not a proven mechanism: a direct causal test (e.g. ablating the visual stream mid-rollout) is left for follow-up.

5.2 Why this, not alternative accounts

Four alternative explanations are ruled out: skill difference (success bunched 93–99%), probe capacity (DtG decodes everywhere), sample-size or stochastic-sampling artefacts (Appendix A), and linear-probe artefact (memory transplant reproduces the ordering outside the probe framework; MLP probe shows the divergence is specific to *linear* top-layer decodability, which is what the linear DD-PPO action head reads). Full rebuttals in Appendix C.

5.3 Biological precedent

Each of our three findings has a direct parallel in animal-navigation research: compensatory hippocampal coupling in congenitally blind humans [Kupers and Ptito, 2014]; grid-cell periodicity collapse under darkness with navigation preserved [Chen et al., 2016]; cross-modality hippocampal remapping in bats [Geva-Sagiv et al., 2016]; the primate vs. rodent view-cell/place-cell contrast attributed to foveated vs. near-panoramic sampling [Rolls, 2024]; and saccade-gated hippocampal theta whose reliability predicts memory encoding [Jutras et al., 2013, Tas et al., 2016]. We use these as convergent (not conclusive) evidence that the effects are not specific to a single LSTM architecture. The parallels make a pure single-architecture artefact less likely; we do not claim the agent and biological mechanisms are the same.

5.4 Implications for ML practice and architectural scope

Within the recurrent PointNav setting we study, three implications follow. (i) *Encoder capacity correlates inversely with linear, top-layer, policy-accessible LSTM spatial encoding*; if causal (Appendix H tests this), it is also a practical training-time lever for inducing cognitive-map-like internal representations. (ii) *"More pixels" is not the right lens*: the linear top-layer GPS code is non-monotonic in raw pixel count (uniform 256^2 lacks it; matched 48^2 with 1×1 feature map has a strong one). The encoder feature-map probe shows neither encoder linearly preserves GPS, so the mechanism is not encoder position retention; we attribute the pattern to encoder spatial-feature variety per step, with the scaling sweep and log-polar control needed to confirm. (iii) *Gaze (H3 pending):* within the rich-encoder regime, we predict shifting the static gaze location alters what the LSTM encodes; the foveated-shifted causal control is in training (§4.5).

Beyond the recurrent setting (untested) The three implications are established only in our LSTM setting. Transformer-based embodied agents build their internal state by accumulating information via learned attention, structurally similar to our gaze location in the sense that it selects which observation content is read into the representation; H3 therefore suggests a testable prediction for transformer navigators (strong constraints on attended tokens $\rightarrow$ content-level dissociations like ours) which we do not verify here. The sensor-structure axis (H1, H2) is a claim specifically about RL recurrent navigation agents under partial observability; whether equivalent effects arise in supervised visual learners, non-visual modalities, or multimodal systems is an open question.

5.5 Limitations

(i) *Intermediate checkpoint for foveated-fixed:* all foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the silent-corruption window (Appendix F); the other four conditions use the final converged checkpoint. (ii) *Minimal learned-gaze architecture.* Our learned-gaze module is a deterministic sigmoid MLP with slow-gaze rollout updates, chosen for computational compatibility with DD-PPO rollouts and simplicity as a controlled variable, not as a competitive active-gaze model. Its collapse under pure task supervision is specific to this architecture (and so far to a single seed; multi-seed replication is in progress); richer gaze mechanisms (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. We therefore do not claim that implicit task pressure cannot produce active gaze in general; we claim only that our minimal decoder collapsed, and we use the collapsed-gaze position as an empirical starting point for the gaze-location test (foveated-shifted control, §4.5) — a test whose logic does not depend on the gaze decoder working. (iii) *Cross-heading probe:* the proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes and does not appear in our results. (iv) *Foveation model:* Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. A log-polar control (F3) and a $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep (F1–F4) are in training at submission and will populate §4.4 and Appendix G. (v) *Linear probes only:* non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. (vi) *Architecture scope:* we use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze location as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 for the specific predictions.

6 Conclusion

We ran a controlled five-condition ablation isolating the effect of visual-input structure on the recurrent representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge.

(1) Encoder–memory race (H1). Top-layer LSTM GPS encoding is *temporally stable* when the encoder hands the LSTM minimal spatial-feature variety per step (blind, matched-compute), and *transient* when richer visual features substitute (uniform, foveated, foveated-learned). The encoder feature-map probe shows no sighted encoder linearly decodes GPS, so the H1 mechanism lives inside the LSTM, not at the encoder. H1 is most precisely a claim about *linear, top-layer, late-episode* GPS encoding — what the linear DD-PPO action head can read.

(2) Condition-specific representational format (H2). Even where rich-encoder LSTMs are spatially pass-through, their hidden-state subspaces are linearly separable to single-nearest-neighbour granularity (CKA $< 10^{-4}$, probe transfer $\ll -800$, 1-NN purity 1.000). The 5×5 memory-transplant matrix is asymmetric: bottleneck donor states hurt rich-encoder recipients far more than the reverse. Whatever non-spatial task-relevant features each LSTM carries, it carries them in its own linear subspace.

(3) Gaze location (H3, pending) and a having–vs.–using dissociation. Whether gaze location is a second content axis *within* the rich-encoder regime is in training (foveated-shifted control). Independently of that pending test, plotting probe-readable GPS against behavioural memory reliance reveals a 2×2 dissociation: matched has linear GPS but ignores it; uniform has no linear GPS but heavily uses memory. Both anomalies rest on single-seed runs and we mark them as candidate dissociations pending multi-seed replication.

The pattern parallels three threads of animal-navigation research — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in darkness with navigation nonetheless preserved, and cross-modality hippocampal remapping in bats. We use these as analogy only: the convergent direction across systems is suggestive of a sensor–memory coupling principle, but our experiments do not test architectures other than LSTM, so we do not claim the principle is architecture-independent. Open questions: (i) whether the encoder-bottleneck ordering extends to finer-grained resolution sweeps (Appendix H); (ii) whether H3 stands on the clean causal test; (iii) whether transformer-based navigation agents reproduce the encoder–memory race; (iv) whether equivalent effects shape representations in non-navigation embodied tasks; (v) multi-seed replication of all five conditions to bound seed-to-seed variability of the H1 numbers.

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A Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.]**

Cross-heading probe The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

B Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states + GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, foveated-learned 0.73, uniform 0.55 (vs. Ridge's chance R^2). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the resolution scaling sweep (Appendix H) tests this more directly by varying input resolution at fixed encoder stack.

C Alternative-account rebuttals (full)

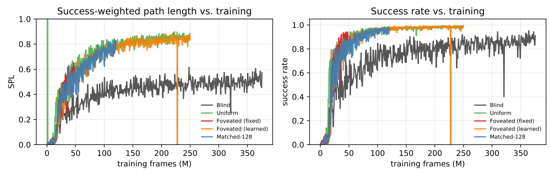
Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them. Hewitt–Liang label-permutation selectivity is high where GPS is high (blind, matched) and at chance where GPS is at chance — the latter follows from real and control probes facing the same near-zero target-information ceiling. The chance-level rich-encoder R^2 is therefore a hidden-state property, not a probe artefact.

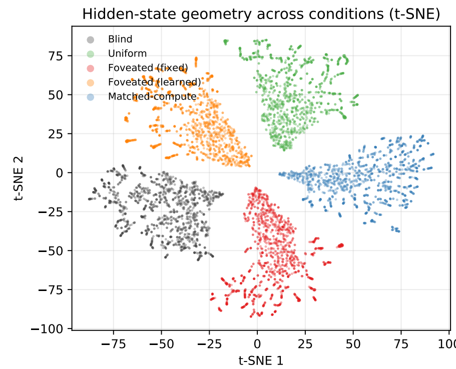
Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix A.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering; the 5×5 matrix's asymmetry adds a structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix B). The bottleneck-vs-pass-through divergence is therefore not about whether *any* probe can recover position; it is specifically about whether a *linear* top-layer probe can. Since the DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear metric is what the policy itself can use, so H1 maps onto policy-accessible encoding regardless of how the MLP interpretation resolves.

D Supplementary visualisations



(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all five conditions (perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 7: Supplementary visualisations referenced from the main text.

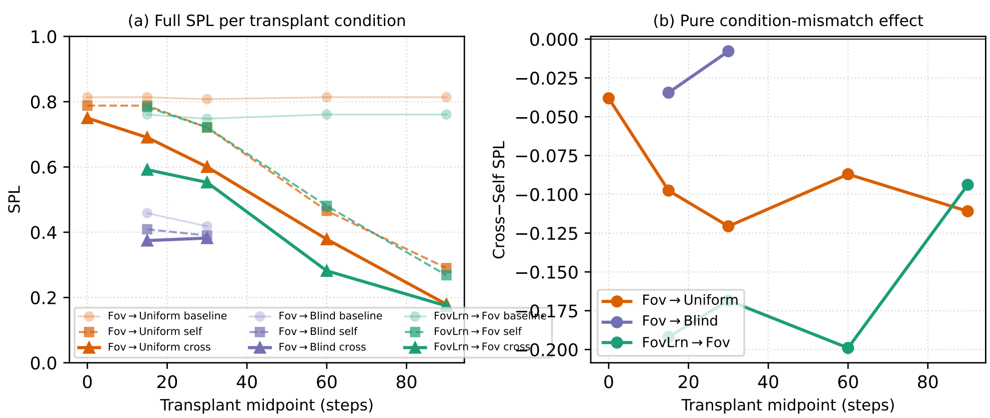


Figure 8: Memory-transplant midpoint sweep on three representative donor$\to$recipient pairs (foveated$\to$uniform, foveated$\to$blind, foveated-learned$\to$foveated) across midpoints $\{0, 15, 30, 60, 90\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in Figure 3 (right) is at midpoint t=30; this supplementary figure shows that the cross-self gap is roughly stable for $t \geq 15$ and that the t=0 self-/cross- baseline collapses to a protocol-noise floor (so the diagonal in Figure 3 is correctly read as "rollout-divergence noise" rather than "condition-mismatch noise"). Foveated-learned$\to$foveated incurs -0.17 SPL on average across $t \geq 15$, foveated$\to$uniform -0.10, foveated$\to$blind -0.02, consistent with the asymmetry pattern in the full matrix.

E Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0,1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears

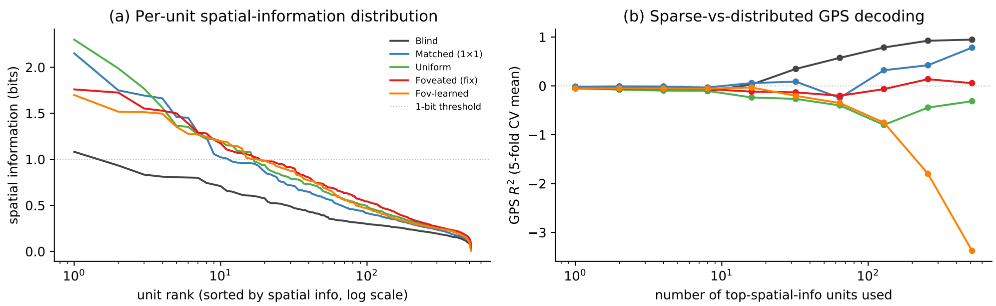


Figure 9: Population coding analysis (referenced from §4.6). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / fov-learned) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; matched climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

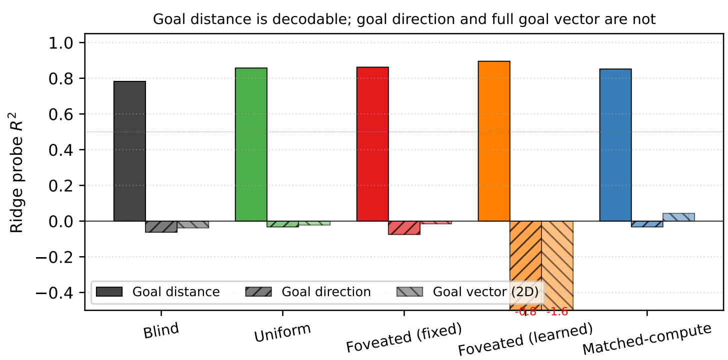


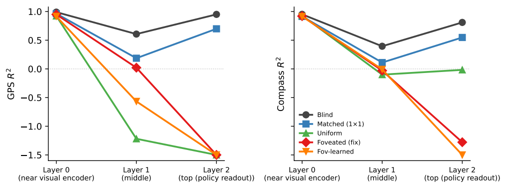
Figure 10: Goal-vector probe R^2 (referenced from §4.6). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \mathrm{Var}[g]$ variant as a negative control.

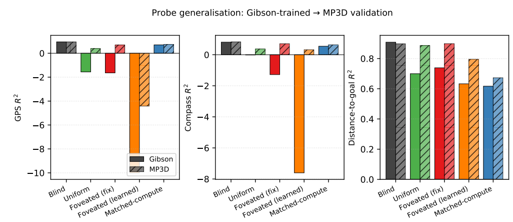
Pilot 1 (uncapped, job 2840256) After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288) Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all five conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 2d (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-pass-through partition is the same.

Figure 11: Per-layer and MP3D companion figures (referenced from §4.2).

entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

F Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$= 0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

G Foveation experiments: status and design

This appendix documents the four controlled foveation experiments described in §4.4. All four were submitted to the cluster at the time of writing; the table below tracks their state. Numbers will populate the corresponding paragraphs in §4.4 in revision.

Table 4: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe (matched / uniform / foveated)	where position information is held	in probing
Foveated-shifted (gaze $(0.49, 0.62)$)	gaze-location axis (links to H3)	in training

Why disclose this rather than wait. The four experiments are designed to falsify or refine specific predictions of the encoder–memory race framework, and each has a falsifiable signature. We disclose the design openly so that (i) the present submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) any reader can identify which result would change the paper's interpretation. The current evidence already places foveation-fixed and foveation-learned within the framework; the experiments above sharpen the "foveation" part of the story specifically.

H Encoder-capacity scaling sweep

[TODO: The scaling sweep trains matched-compute variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 12 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256, crossing into the pass-through regime between 48 and 128 pixels.]

Figure 12: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean$\pm\sigma$).